Overview

Applied statistics in ML means making valid decisions from model outputs: choosing metrics, diagnosing error modes, and aligning evaluation with business cost.

Correlation vs Causation

- Correlation: statistical association.
- Causation: intervention on one variable changes another.

Predictive models can be useful without causal interpretation, but decision-making must respect this limitation.

Bias/Variance & Overfitting/Underfitting

Expected prediction error decomposes into bias, variance, and irreducible noise (conceptually).

- High bias: model too simple, systematic error.
- High variance: model too sensitive to training fluctuations.
- Underfitting: high bias.
- Overfitting: high variance.

Evaluation Metrics

Classification

- Accuracy: $
  rac{TP+TN}{N}$
- Precision: $
  rac{TP}{TP+FP}$
- Recall: $
  rac{TP}{TP+FN}$
- F1: harmonic mean of precision and recall.
- ROC-AUC: ranking quality across thresholds.

Regression

- MAE, MSE, RMSE, R^2.

Metric choice must match error cost profile.

Iris Classification Example

Iris offers multiclass baseline to compare metrics and confusion patterns. Use train/test split and standardized features before logistic regression.

Common Pitfalls

- Reporting only accuracy for imbalanced problems.
- Tuning on test set (leakage).
- Ignoring calibration when probabilities drive decisions.

Business Use Cases

- Fraud detection: recall often prioritized.
- Medical triage: high recall with controlled precision.
- Marketing targeting: precision-sensitive to avoid wasted spend.

Business Case Studies & Exceptions

Case 1: Accuracy-Led Decision Failure

Model with 95% accuracy missed rare but costly fraud cases.

Fix: - Shift optimization to recall/F1/PR-AUC. - Use class weighting and threshold tuning.

Case 2: Poor Calibration in Risk Scores

Predicted 0.9 risk did not correspond to actual event rate, causing bad capacity planning.

Fix: - Add calibration diagnostics. - Use Platt scaling or isotonic calibration if needed.

Interview Questions & Answers

1. **Q: Precision vs recall?**
   A: Precision measures false-positive control; recall measures false-negative control.
2. **Q: When use F1?**
   A: When balancing precision and recall under class imbalance.
3. **Q: Why ROC-AUC can mislead?**
   A: It may look strong even when precision in rare-positive region is poor.
4. **Q: What is data leakage?**
   A: Information from validation/test leaks into training, inflating metrics.
5. **Q: Bias-variance tradeoff in practice?**
   A: Increasing model complexity often reduces bias but increases variance.
6. **Q: Why calibration matters?**
   A: Business decisions based on probability thresholds require trustworthy probabilities.
7. **Q: Why keep separate train/val/test sets?**
   A: To tune, select, and estimate generalization without optimistic bias.
8. **Q: Threshold tuning example?**
   A: Lower threshold to catch more fraud (higher recall) at cost of more false alerts.
9. **Q: Metric for symmetric regression error?**
   A: RMSE or MAE depending outlier sensitivity preference.
10. **Q: What is good evaluation report?**
    A: Multiple metrics, confusion matrix, segment breakdown, and assumptions/limitations.